

RG-Improved Transient Matching and Explicit Thermal Collision Kernel

Angus Muffatti

Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral; see RECOVERY_PROVENANCE.md.

48. RG-improved hard function

The fixed-order function was

$$D_{FFS}^{\text{fixed}}(\bar{\mu}) = 2 \ln \frac{x}{z} \ln \frac{x-z}{\bar{\mu}^2} - \ln^2 \frac{x}{\bar{\mu}^2} - 2 \text{Li}_2\left(\frac{z}{x}\right) + \frac{\pi^2}{3}.$$

Combining it with operator running and the matching counterterm yields

$$D_{FFS}^{\text{hard}} = 2 \ln \frac{x}{z} \ln \left(1 - \frac{z}{x}\right) - 2 \text{Li}_2\left(\frac{z}{x}\right) + \frac{\pi^2}{3}.$$

Its explicit matching-scale derivative vanishes. The benchmark remains

$$j_3^{\text{hard}} = 6.57973508149.$$

Ordinary coupling and portal running leaves only

$$0.96469 < \frac{C_3(\mu_D)}{C_3(M)} < 1.03745$$

for $M/2 < \mu_D < 2M$.

Restoring the full Higgs doublet and colour multiplicities gives

$$C_3^{\text{full}} = 1.58896 \times 10^{-4} \text{ GeV}^2,$$

$$|\Delta V_{Q_a}^{(3)}| = 4.2902 \times 10^3 \text{ GeV}^4,$$

still only 3.053×10^{-12} of the thermal potential.

A separate warning survives: in the minimal one-loop high-scale run, the Higgs quartic becomes negative. The matching calculation is stable, but the displayed scalar sector is not yet a complete metastable UV benchmark.

49. Reduced explicit collision kernel

The v1.3 calculation replaced a fitted BGK closure by a microscopic discrete gain/loss operator for

$$H, \quad D, \quad q, \quad g.$$

It included six $1 \leftrightarrow 2$ channel families, eight $2 \leftrightarrow 2$ families, Bose enhancement, Pauli blocking, Debye screening, asymptotic thermal masses, an LPM formation-time reduction, and exact discrete energy conservation.

The lattice contained 514 collinear transitions and 14,289 elastic transitions. It reached 99% of the equilibrium entropy gain by $Tt = 3920$, with detailed-balance residual 8.57×10^{-17} and energy drift 1.44×10^{-15} . The resulting hierarchy was

$$\frac{\Gamma_{\text{kin}}^{(99)}}{\Gamma_R} = 1.729 \times 10^6.$$

Therefore homogeneous energy bookkeeping can adiabatically eliminate the fast plasma without a phenomenological BGK rate [AMY2002; Boedeker2019].